

lect #3

$$X, \mathcal{E} \subset \mathcal{P}(X) : \phi \in \mathcal{E}, X = \bigcup_{n=1}^{\infty} X_n, X_n \in \mathcal{E}$$

$$g: \mathcal{E} \rightarrow [0, +\infty], g(\emptyset) = 0$$

$$\mu^*: \mathcal{P}(X) \rightarrow [0, +\infty], \\ A \subseteq X, \mu^*(A) := \inf \left\{ \sum_{n=1}^{\infty} g(E_n) : A \subseteq \bigcup_{n=1}^{\infty} E_n, E_n \in \mathcal{E} \right\}$$

Prop 2.7: μ^* outer measure

$$\mu^*(E) \leq g(E), E \in \mathcal{E}$$

Corollary 2.8: $(\mathbb{R}^N)^*$, μ^*_f , H^s (defined in 2.2, 2.3, 2.4) are outer measures

Question: ? $g(E) = \mu^*(E)$ if $E \in \mathcal{E}$?

Proposition 2.9 $\mu^*(E) = g(E) \forall E \in \mathcal{E}$ iff g is countably subadditive

i.e. $g(E) \leq \sum_{n=1}^{\infty} g(E_n)$
 whenever $E, E_n \in \mathcal{E}$ and $E \subseteq \bigcup_{n=1}^{\infty} E_n$.

Proof [$\Leftarrow$] Assume that g is countably subadditive.
 If $E \subseteq \bigcup_{n=1}^{\infty} E_n, n \in \mathbb{N}$ then

$$g(E) \leq \sum_{n=1}^{\infty} g(E_n).$$

Self on RHS are all such coverings, get

$$g(E) \leq \mu^*(E).$$

Thus + Prop 2.7 $\Rightarrow g(E) = \mu^*(E)$

[$\Rightarrow$] $E, E_n \in \mathcal{E}, E \subseteq \bigcup_{n=1}^{\infty} E_n$
 $g(E) = \mu^*(E) \leq \sum_{n=1}^{\infty} g(E_n)$
 (def of μ^*)

hyp

countably subadditive.

Prop 2.10 $(\mathbb{I}_e^0)^*(R) = \inf \{ \sum_{i=1}^n l(R_i) : R \subseteq \bigcup_{i=1}^n R_i, R_i \text{ open} \} = \alpha(R)$
 $\uparrow$ not nec/open/closed.

$$l([a,b]) = l((a,b)) = l([a,b]) = l([a,b])$$

Prop 2.11 $(\mathbb{I}_e^0)^*(R) = l(R) \quad \forall R \text{ rectangle}$

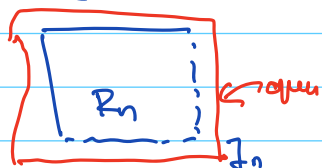
Proof Always (Prop 2.7) $(\mathbb{I}_e^0)^*(R) \leq p(R)$.

Step 1 claim that $(\mathbb{I}_e^0)^*(R) = \inf \{ \sum_{i=1}^n l(R_i) : R \subseteq \bigcup_{i=1}^n R_i, R_i \text{ open} \} = \alpha(R)$

$$(\mathbb{I}_e^0)^*(R) \leq \alpha(R) \text{ clearly because we are restricting the admissible sets on RHS}$$

Fix $\epsilon > 0$, let R_n rectangles (not nec. open)
 $\sum_{n=1}^{\infty} l(R_n) \leq (\mathbb{I}_e^0)^*(R) + \epsilon, \quad R \subseteq \bigcup_{n=1}^{\infty} R_n$

$\forall$ Dilate slightly each rect



$$J_n \supset R_n, J_n \text{ open} \\ l(J_n) \leq l(R_n) + \frac{\epsilon}{2^n}$$

Then $R \subseteq \bigcup_{n=1}^{\infty} J_n$

$$\begin{aligned} (\mathbb{I}_e^0)^*(R) &\leq \sum_{n=1}^{\infty} l(J_n) \leq \sum_{n=1}^{\infty} l(R_n) + \epsilon \\ &\leq (\mathbb{I}_e^0)^*(R) + 2\epsilon \end{aligned}$$

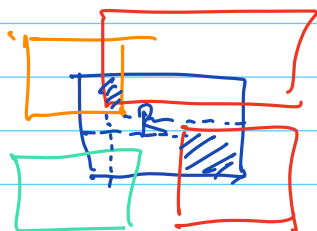
let $\epsilon \rightarrow 0^+$.

Step 2 R is a closed rectangle
claim $l(R) \leq (\mathbb{I}_e^0)^*(R)$

Fix $\epsilon > 0$. Step 1: $\exists R_n \xrightarrow{(\cdot)}$, $R \subseteq \bigcup_{n=1}^{\infty} R_n$

$$\sum_{n=1}^{\infty} l(R_n) \leq (\mathbb{I}_e^0)^*(R) + \epsilon.$$

$$R \text{ compact} \Rightarrow \exists n \in \mathbb{N} : R \subseteq \bigcup_{i=1}^n R_i.$$



Extend the faces of each R_n to hyperplanes in order to get a finite partition of R ... subrectangles J_k

$$\begin{aligned} l(R) &= \sum_{i=1}^k l(J_k) \leq \sum_{i=1}^n l(R_i) \leq \sum_{i=1}^{\infty} l(R_i) \\ &\leq (l^N)^*(R) + \varepsilon \end{aligned}$$

let $\varepsilon \rightarrow 0^+$ get

$$l(R) \leq (l^N)^*(R)$$

Step 3 R arbitrary

Fix $\varepsilon > 0$, find $J \subset R$ closed rectangle
 $l(R) \leq l(J) + \varepsilon$



$$\begin{aligned} l(R) &\leq l(J) + \varepsilon \leq (\text{Step 2}) (l^N)^*(J) + \varepsilon \\ &\quad \uparrow \text{closed} \\ &\leq (\text{monotonicity}) (l^N)^*(R) + \varepsilon. \end{aligned}$$

$$\text{let } \varepsilon \rightarrow 0^+ : l(R) \leq (l^N)^*(R).$$

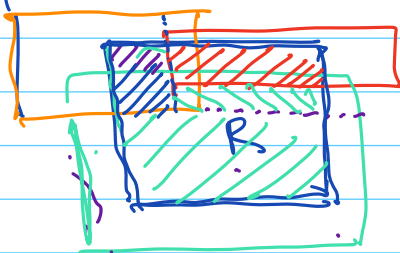
($\geq$ always)

Prop 9.12 $(l^N)^*$ is translation invariant

$$l(x + I) = l(I) \quad x \in \mathbb{R}, I \text{ interval.}$$

$$l(R) = l(x_0 + R), \quad x_0 \in \mathbb{R}^N, R \subset \mathbb{R}^N$$

Back to partition



4

would like: additivity $A \cup B = \emptyset \Rightarrow (\mathbb{L}^n)^*(A \cup B) = (\mathbb{L}^n)^*(A) + (\mathbb{L}^n)^*(B)$

Falte (in general)!

Problem: $(\mathbb{L}^n)^*(A)$ is defined for any $A \subseteq \mathbb{R}^n$

will remain some subsets of $\mathbb{R}^n$ ("bad") no measure!

Prmk 2.13 we construct $A, B \subseteq \mathbb{R}$ such that
 $A \cap B = \emptyset$ and $(\mathbb{L}^1)^*(A \cup B) < (\mathbb{L}^1)^*(A) + (\mathbb{L}^1)^*(B)$

[1,1] consider the equivalence relation:
 $a \sim b$ iff $a - b \in \mathbb{Q}$

$\forall a \in [1,1], a \in [a] := \{b \in [1,1] : a \sim b\}$

$$[1,1] = \bigcup_{a \in [1,1]} [a]$$

Axiom of Choice: If $\mathcal{E}$ is a set whose elements are disjoint,

nonempty subsets of X , then there exists $\forall x \in X$ that contains exactly one element of each element of $\mathcal{E}$.

Take $\mathcal{E} := \{[a] : a \in [1,1]\}$

↓

$V \subseteq \mathbb{R}$, V contains exactly one element of each equivalence class:

$V \cap [a]$ has only one element.

$[1,1] \cap \mathbb{Q} = \{r_1, r_2, r_3, \dots, r_n, \dots\}$ distinct rational numbers.

claim $[1,1] \subset \bigcup_{k=1}^{\infty} (r_k + V)$

why: $a \in [1,1]$ let $V \cap [a] = \{\tilde{a}\}$

$$a - \tilde{a} \in \mathcal{O} \Rightarrow a = \tilde{a} + r_k, r_k \in \mathbb{N} \quad (\text{note } a - \tilde{a} \in [-2, 2])$$

Subadditivity of the outer measure + Prop 2.11:

$$2 = \lambda([-1, 1]) = \underbrace{(\mathbb{Q}^1)^*([-1, 1])}_{\text{Prop 2.11}} \leq \sum_{k=1}^{\infty} (\mathbb{Q}^1)^*(r_k + V)$$

$$(\text{Transl. invariant}) := \sum_{k=1}^{\infty} (\mathbb{Q}^1)^*(V)$$

$$\Rightarrow (\mathbb{Q}^1)^*(V) \geq 0$$

claim $(r_j + V) \cap (r_i + V) = \emptyset$ if $i \neq j$ (1)

why: Suppose $\exists i, j, i \neq j, t \in (r_i + V) \cap (r_j + V)$ some $t \in \mathbb{R}$

$$t = r_j + V_j = r_i + V_i, \text{ some } v_i, v_j \in V$$

$$\Rightarrow V_i - V_j = \underbrace{r_j - r_i}_{\in \mathbb{Q}} \Rightarrow [V_i] = [V_j]$$

$$\Rightarrow V_i = V_j \Rightarrow r_i = r_j \Rightarrow i = j.$$

now $V \subseteq [-1, 1], r_k \in [-2, 2] \forall k$

$$\bigcup_{k=1}^n (r_k + V) \subseteq [-3, 3]$$

Monotonicity of $(\mathbb{Q}^1)^*$: $(\mathbb{Q}^1)^*\left(\bigcup_{k=1}^n (r_k + V)\right) \leq (\mathbb{Q}^1)^*([-3, 3])$
 $= \lambda([-3, 3]) = 6$ (2)

Translation invariance:

$$\sum_{k=1}^n (\mathbb{Q}^1)^*(r_k + V) = \sum_{k=1}^n (\mathbb{Q}^1)^*(V) = n \underbrace{(\mathbb{Q}^1)^*(V)}_{>0}$$

choose $n > 1$ s.t. $n (\mathbb{Q}^1)^*(V) > 6$ (3)

If $(\mathcal{I}_\epsilon)^*$ was finite additive,

$$\textcircled{6} \stackrel{\textcircled{2}}{=} (\mathcal{I}_\epsilon)^* \left(\bigcup_{k=1}^n (r_k + V) \right) \stackrel{\textcircled{1}}{=} \sum_{k=1}^n (\mathcal{I}_\epsilon)^* (r_k + V) \quad (\mathcal{I}_\epsilon)^* (V)$$

$$(\text{trans inv}) = n (\mathcal{I}_\epsilon)^* (V)$$

$$\stackrel{\textcircled{3}}{> 6}$$

$$A_i = r_i + V, \quad B_i = \bigcup_{k=2}^i (r_k + V).$$

Can prove:

Prop 2.14 $\exists \mu: \mathcal{P}(\mathbb{R}) \rightarrow [0, +\infty] \leq 1.$

$$(i) \quad \mu(I) = l(I) \quad \forall I \text{ interval in } \mathbb{R}$$

$$(ii) \quad \mu \left(\bigcup_{n=1}^{\infty} A_n \right) = \sum_{n=1}^{\infty} \mu(A_n) \text{ whenever } \{A_n\} \text{ mutually disjoint}$$

$$(iii) \quad \mu(t + A) = \mu(A) \quad \forall t \in \mathbb{R}$$